WORKED SOLUTION

STEP III 2015, Question 2

Pure • Analysis

STEP III 2015, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q2

This question has a full guided treatment in *Beyond STEP*, Section 2.1, *What "Eventually" Means*: https://beyondstep.keithmcnulty.org/ch02/eventually.html

The key idea

The relation $(s_n) \leq (t_n)$ says "eventually $s_n \leq t_n$": we may ignore any finite number of early terms. So a true statement needs a proof that names a point m beyond which the inequality is guaranteed, and a false one needs a pair of sequences that defeat every possible choice of m. Keep asking: *from which n onwards* does this work?

Solution

Part (i)

True. For a positive integer n we can divide by $n > 0$, so

$$1000n \leq n^2 \iff 1000 \leq n.$$

Hence $1000n \leq n^2$ whenever $n \geq 1000$, and $m = 1000$ works. (No smaller m does, since $1000 \cdot 999 > 999^2$.)

Part (ii)

False. The trap is that "not eventually $s_n \leq t_n$" only tells us that $s_n > t_n$ *infinitely often*, not *eventually*. Two sequences can keep overtaking each other. Take

$$s_n = 2 + (-1)^n, \qquad t_n = 2,$$

so s_n runs $1, 3, 1, 3, \ldots$ and every term of both sequences is positive.

- It is not true that $(s_n) \leq (t_n)$: whatever positive integer m is proposed, there is an even $n \geq m$, and for it $s_n = 3 > 2 = t_n$.

- It is not true that $(t_n) \leq (s_n)$ either: whatever m is proposed, there is an odd $n \geq m$, and for it $t_n = 2 > 1 = s_n$.

So the hypothesis of the statement holds but its conclusion fails.

Part (iii)

True. By hypothesis there is a positive integer m_1 with $s_n \leq t_n$ for all $n \geq m_1$, and a positive integer m_2 with $t_n \leq u_n$ for all $n \geq m_2$. Let

$$m = \max(m_1, m_2).$$

If $n \geq m$ then $n \geq m_1$ and $n \geq m_2$, so both inequalities hold at n, giving

$$s_n \leq t_n \leq u_n.$$

Thus $s_n \leq u_n$ whenever $n \geq m$, which is $(s_n) \leq (u_n)$.

Ask yourself Why not simply take $m = m_1$, or $m = m_1 + m_2$?

Taking $m = m_1$ fails because for $m_1 \leq n < m_2$ we know nothing about t_n versus u_n. We need n to be past *both* thresholds at once, and $\max(m_1, m_2)$ is the smallest choice that guarantees this. ($m_1 + m_2$ also works, but it is wasteful.)

Part (iv)

True, with $m = 4$. A short table shows where to start:

n	1	2	3	4	5	6
n^2	1	4	9	16	25	36
2^n	2	4	8	16	32	64

The inequality holds at $n = 1, 2$, fails at $n = 3$ and holds again from $n = 4$. So $m = 4$ is the smallest possible choice, and we prove by induction that $n^2 \leq 2^n$ for all $n \geq 4$.

Base case. At $n = 4$, $n^2 = 16 = 2^4$.

Inductive step. Suppose $n \geq 4$ and $n^2 \leq 2^n$. Since $n \geq 3$ we have $n^2 - 2n - 1 = (n-1)^2 - 2 \geq 2 > 0$, so $2n + 1 < n^2$. Therefore

$$\begin{aligned}
(n+1)^2 &= n^2 + 2n + 1 \\
&< n^2 + n^2 && \text{since } 2n + 1 < n^2 \\
&= 2n^2 \\
&\leq 2 \cdot 2^n && \text{by the inductive hypothesis} \\
&= 2^{n+1}.
\end{aligned}$$

So the result for n implies it for $n + 1$, and by induction $n^2 \leq 2^n$ for every $n \geq 4$. Hence $(n^2) \leq (2^n)$ with $m = 4$.

Ask yourself Where exactly did the induction need n to be large?

Only in the step $2n+1 < n^2$, which needs $n \geq 3$. That is why the induction cannot run from $n = 1$: the step from $n = 2$ to $n = 3$ uses $5 < 4$, which is false, and indeed $3^2 > 2^3$.

Checks

- Part (i): at $n = 1000$ both sides equal 10^6; at $n = 999$, $999\,000 > 998\,001$.
- Part (iv): checking $n^2 \leq 2^n$ by computer for every n from 4 to 2000 confirms it, and the table above confirms the failure at $n = 3$. For a feel of how fast 2^n wins: $20^2 = 400$ but $2^{20} = 1\,048\,576$.
- Part (ii): any m you try is beaten by the next even n (for one order) and the next odd n (for the other).

Common slips

- Treating "not eventually $\leq$" as "eventually $>$". The negation of "there is an m such that for all $n \geq m\ldots$" is "for every m there is some $n \geq m$ with $s_n > t_n$".
- Starting the induction in (iv) at $n = 1$ or $n = 2$: the inductive step genuinely fails there.
- In (iii), using the same letter m for both hypotheses, as though the two thresholds had to coincide.

Beyond STEP

The relation here is the "eventually" quantifier that runs through all of real analysis: a sequence converges if, for every tolerance, it is *eventually* within that tolerance of its limit. Part (ii) shows that, unlike the order on real numbers, this ordering of sequences is not total: two sequences need not be comparable at all. Computer scientists use the same idea for big-O notation, where (iv) is the familiar fact that exponentials eventually beat polynomials. Abbott's *Understanding Analysis* is a friendly place to see these quantifier arguments made systematic.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.